

Browser fingerprinting

Local Flask prototype | 2 October 2026

The aim was to see what a site can learn without placing a cookie. I used a Flask server on **127.0.0.1:8000**, then opened it in a Chromium-based browser and Safari on the same Mac. The page records HTTP headers, asks JavaScript for browser features, and measures one typing attempt.

Run the project

```
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
python app.py
```

1. Passive clues in the HTTP request

The server prints these values when the page is requested. JavaScript has not run at this point. I added six request headers beyond User-Agent and Accept-Language; a missing header is shown as *(not sent)*.

Header	What it reveals in this test
Accept	The response formats the browser accepts, including HTML and image formats.
Accept-Encoding	Compression formats: Chromium sent gzip, deflate, br, zstd; Safari sent gzip, deflate.
Referer	Navigation source. It was absent because I opened the address directly.
Sec-Fetch-Site / Mode	Request context. Both browsers sent none / navigate for the direct page load.
Sec-CH-UA	Browser brands and versions. Chromium sent it; Safari did not.

```
===== NEW VISIT =====
IP address: 127.0.0.1
HTTP method: GET
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like
Gecko) Chrome/154.0.0.0 Safari/537.36
Accept-Language: en-US,en;q=0.9
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/
png,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate, br, zstd
Referer: (not sent)
Sec-Fetch-Site: none
Sec-Fetch-Mode: navigate
Sec-CH-UA: "Chromium";v="154", "Google Chrome";v="154", "Not A(Brand";v="99"
=====
```

Figure 1. Flask output from the initial page request; no JavaScript was needed.

The IP address in this local test was 127.0.0.1. It is the loopback address, not a public address for the person using the browser.

2. Active browser features

The page script reads browser APIs after loading and sends a JSON object to **/collect**. These values add detail that the original page request did not necessarily contain.

Object	Feature and why it matters
navigator	Language and logical core count describe browser settings and reported hardware.
screen	Screen size and colour depth describe the display environment.
window	Window size and pixel ratio describe the current viewport and display scale.
Intl	The time zone reflects regional configuration.
document	Referrer describes the previous page when one is supplied.

Browser Fingerprinting Lab

Local experiment: this page shows what the browser exposes without cookies.

Active browser features

- language: en-US
- logicalCores: 15
- screenSize: 1512 x 982
- colorDepth: 30
- windowSize: 847 x 853
- pixelRatio: 2
- timeZone: Africa/Casablanca
- referrer: (direct visit)

SHA-256 fingerprint: cc499565729c1b9177647302ef7636c734d7df385680caee918e72076d5e2038

Typing rhythm

Type this sentence exactly:

Cookies are gone, but Alice's rhythm still leaves traces!

Restart

Figure 2. Values displayed by the Chromium-based browser, with its SHA-256 result.

This run reported en-US, 15 logical cores, a 1512 x 982 screen, 30-bit colour depth, and Africa/Casablanca. Flask received the same feature values in a POST request. A browser may omit or reduce any of these fields, so they are clues rather than proof of identity.

3. Typing behaviour

The timer begins on the first printable key. A keydown listener counts Backspace presses; an input listener checks for an exact match with the displayed sentence. On a match, the script displays the measurements and posts them to Flask.

Formula: typing speed = number of target characters / elapsed seconds. The sentence has 57 characters, including spaces and punctuation. In the run below, $57 / 16.23 = 3.51$ characters per second.

Algorithm: clear state on Restart; set the start time on the first character; count Backspace after timing begins; on an exact match, stop, calculate, display, and send the three measurements. The input converts a straight keyboard apostrophe to the curly apostrophe printed in the target sentence.

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Cookies are gone, but Alice's rhythm still leaves traces!

Restart

Time: 16.23 s | Speed: 3.51 characters/s | Backspace corrections: 4

Figure 3. Completed typing run: 16.23 s, 3.51 characters/s, four Backspace presses.

A later run reached /collect with 3.22 s, 17.68 characters/s and zero corrections. The difference between two runs on one person shows why typing rhythm is only a probabilistic signal.

4. Comparison and one hash

For the optional fingerprint, the script puts language, screen size, colour depth, pixel ratio, time zone and logical core count in a fixed order, then applies SHA-256. Typing is excluded so the hash can be checked across reloads. The Chromium hash stayed the same on three loads; Safari produced a different hash.

Feature	Chromium-based browser	Safari
Language	en-US	en-GB
Logical cores	15	8
Screen size	1512 x 982	1512 x 982
Colour depth	30	24
Time zone	Africa/Casablanca	Africa/Casablanca
Sec-CH-UA	Present	Not sent

Browser Fingerprinting Lab

Local experiment: this page shows what the browser exposes without cookies.

Active browser features

- language: en-GB
- logicalCores: 8
- screenSize: 1512 x 982
- colorDepth: 24
- windowSize: 1512 x 807
- pixelRatio: 2
- timeZone: Africa/Casablanca
- referrer: (direct visit)

SHA-256 fingerprint: 6e47b6ef4d4ec755a345ebaf7dd4998ed319fa912924e785ab4b4adba904d8ed

Typing rhythm

Type this sentence exactly:

Cookies are gone, but Alice's rhythm still leaves traces!

Restart

Figure 4. Safari on the same Mac returned a different set of active values.

The shared screen size and time zone survived the browser change; language, reported core count and colour depth did not. Window size also changed because the windows had different dimensions. A hash makes the combination compact, but it does not prove that two visits belong to one person and it does not make the underlying data anonymous.

5. Private Browsing check

I then opened the same local page in Safari Private Browsing. The request still contained the same Safari User-Agent, Accept-Language and Accept-Encoding headers. Most active features also remained visible.

Feature	Safari normal	Safari private
Language / time zone	en-GB / Africa/Casablanca	en-GB / Africa/Casablanca
Logical cores / colour depth	8 / 24	8 / 24
Screen size	1512 x 982	1512 x 845
SHA-256 prefix	6e47b6ef...	67b38975...



Browser Fingerprinting Lab

Local experiment: this page shows what the browser exposes without cookies.

Active browser features

- language: en-GB
- logicalCores: 8
- screenSize: 1512 x 845
- colorDepth: 24
- windowSize: 1512 x 845
- pixelRatio: 2
- timeZone: Africa/Casablanca
- referrer: (direct visit)

SHA-256 fingerprint: 67b389759f7ebf66d038d5235292b9458e599bf6c69d245ddaa0b912043e9e29

Figure 5. The Private label and feature list are crops from the same Safari capture.

Private Browsing changed the reported screen height here, so the hash also changed. A temporary Safari notice reduced the page height again on one reload; after dismissing the notice, the private hash returned to its earlier value. This is a useful limit of the prototype: a size change can look like a new browser even when it is the same one.

Of the modes tested, Safari Private Browsing changed the most visible values compared with normal Safari, but it did not hide language, time zone or reported core count. The test does not establish that private mode prevents fingerprinting. No extension or separate cookie-blocking setting was tested; cookie blocking would target stored identifiers, while this page also reads headers and JavaScript APIs.

References: TP handout; MDN HTTP headers; MDN User-Agent; MDN Sec-Fetch-Site; MDN digest(); Apple Safari Private Browsing; WebKit Tracking Prevention.